

Board of Studies (BoS) Syllabus Template

Course Information

Degree: UG

Program: BCA

Course Name: WEB TECHNOLOGIES

Course Credits: 03

Semester: III

Course Category: Elective (DSE)

Type of Course: Lecture

New Course: No

Learning Hours

Activity	Hours per Week
Lecture	2
Tutorial	0
Practical	2
Experiential Learning	0

Course Contribution

Category	Percentage Contribution
Employment Enhancement	40%
Entrepreneurship Skills	30%
Skill Development	30%

Course Relevance

Level	Percentage Relevance
Global	40%
National	30%
Regional	20%

Local	10%
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Course Objectives (COB)

CO#	Course Objectives
CO1	Identify the fundamental concepts of HTML, CSS, and JavaScript in web development.
CO2	Explain the role of client-side and server-side programming in building dynamic web applications.
CO3	Apply PHP, React, and Node.js to develop interactive and functional web applications.
CO4	Analyze different techniques for handling data, forms, cookies, and sessions in web applications.
CO5	Create a full-stack web application using React, Node.js, and MySQL for real-world use cases.

Course Outcomes (COs)

CO#	Course Outcome	Bloom's Taxonomy Level
CO1	Identify the structure and styling techniques of HTML and CSS for creating web pages.	L2
CO2	Explain the concepts of JavaScript, DOM manipulation, and event handling for interactive web applications.	L2
CO3	Apply PHP and server-side scripting to handle forms, sessions, and cookies.	L3
CO4	Analyze different React.js components and their role in building modern web applications.	L4
CO5	Create a full-stack web application using Node.js, MySQL, and React for database-driven solutions.	L6

SESSION PLAN :

Module	Course Content	Tools Used / Assessment and Activity	Bloom's Level	Mapping COs	Percentage change in Module (put Nil if there is no change or it is a new course write NEW)
MODULE 1 : HTML AND CSS	Form elements, Different input element types	PPT Presentation / Classroom Interaction	BL-2	C01	NIL
	HTML Forms	Live Demo	BL-3	C01	NIL
	Tables, Frames	Hands on activity	BL-3	C01, C03	NIL
	Lists	PPT presentation	BL-3	C01	NIL
	HTML CSS: Introduction, Syntax	PPT presentation	BL-2	C01	NIL
	Internal and external CSS	Live Demo	BL-3	C01	NIL
	Selectors, Styling	Hands on activity	BL-3	C01	NIL
	Pseudo class and elements	PPT presentation	BL-3	C01	NIL
	CSS tables and box models, CSS navigation	Hands on activity	BL-3	C01, C03	NIL
MODULE 2 : CLIENT SIDE PROGRAMMING	JavaScript – Introduction	PPT presentation	BL-1	C02	NIL
	Syntax, Keywords, Data types, Operators	PPT presentation	BL-2	C01	NIL
	Functions, Arrays, Objects	Hands on activity	BL-3	C02	NIL
	Conditional Statements, Loops	Live Demo	BL-3	C01, C03	NIL
	JavaScript DOM Model	PPT presentation	BL-3	C02	NEW

	Exception Handling-Validation	PPT presentation	BL-4	C02	NEW
	DHTML with JavaScript	Hands on activity	BL-5	C02	NEW
	JSON introduction – Syntax	PPT presentation	BL-3	C02	NEW
	Function	Live Demo	BL-3	C02	NEW
	Files	Live Demo	BL-3	C02	NEW
MODULE 3 : SERVER SIDE PROGRAMMING	Introduction to PHP	PPT presentation	BL-2	C03	NIL
	Variables, Data-types, Constants	PPT presentation	BL-2	C03	NIL
	Scope of Variables, Type of Variables	PPT presentation	BL-2	C03	NIL
	Type Casting, Functions	Live Demo	BL-3	C03	NIL
	PHP Server Variables, working with form	Hands on activity	BL-3	C03	NIL
	Uploading files to Web Server using PHP	Hands on activity	BL-4	C03	NIL
	Introduction to session control	PPT presentation	BL-2	C03	NIL
	registering session variables, Destroying the Session variables	PPT presentation	BL-3	C03	NIL
	Cookie, setting cookies with PHP	Hands on activity	BL-2	C03	NIL
	using cookies with sessions, deleting cookies	Hands on activity	BL-3	C03	NIL
MODULE 4 : REACT	Introduction	PPT presentation	BL-2	C04	NEW
	React JSX	PPT presentation	BL-3	C04, C05	NEW

	React JS virtual DOM	Live Demo	BL-3	C04, C05	NEW
	React ReactDOM	Live Demo	BL-3	C04	NEW
	Components, class, events	Hands on activity	BL-4	C04	NEW
	Lists, Forms	Live Demo	BL-3	C04	NEW
	keys, router	Hands on activity	BL-4	C04, C05	NEW
	Rendering Elements	Live Demo	BL-4	C04	NEW
	Conditional Rendering	Live Demo	BL-3	C04	NEW
MODULE 5 : NODE.JS AND MYSQL	Introduction	PPT presentation	BL-2	C05	NIL
	Modules	PPT presentation	BL-3	C01, C05	NIL
	File system	PPT presentation	BL-4	C04,C05	NIL
	Node Package Manager(NPM)	Hands on activity	BL-4	C04,C05	NIL
	Events	Live Demo	BL-4	C05	NIL
	Creating web server, uploading files	Hands on activity	BL-3	C04,C05	NIL
	Node.js MySQL :Create Database	Live Demo	BL-3	C01. C05	NEW
	Create Table-Insert	Live Demo	BL-3	C05	NEW
	Select from, where, order By	Live Demo	BL-3	C01. C05	NEW
	Delete, update, Drop table	Hands on activity	BL-3	C05	NEW
	Limit, join	Hands on activity	BL-3	C01. C05	NEW

Text Books (TB):

TB-1	Robert. W. Sebesta, "Programming the World Wide Web", Eighth Edition, Pearson
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	Education, 2014.
TB-2	Marty Hall and Larry Brown, “Core Web Programming”, Second Edition, Volume I and II, Pearson Education, 2001
TB-3	Robin Nixon, “Learning PHP, MySQL & JavaScript: A Step-by-Step Guide to Creating Dynamic Websites”, 7th Edition, O’Reilly publishers, 2025.
TB-4	Alex Banks , Eve Porcello, “Learning React: Modern Patterns for Developing React Apps” 2nd Edition, O’Reilly publishers, 2020.
TB-5	Prof. Satish Jain (Author), M. Geethalyer, “Web Designing and Publishing”, First Edition, 2020

Reference Books (RB):

RB-1	Brad Dayley, Brendan Dayley, Caleb Dayley, Node.js, MongoDB, and AngularJS Web Development, Pearson Education, 2017 (E-Book)
RB-2	Trachtenberg, Adam, and Sklar, David. PHP Cookbook, O'Reilly Publication, 3rd Edition 2014
RB-3	Nixon R., Learning PHP, MySQL, JavaScript, and CSS: A Step-by-Step Guide to Creating Dynamic Websites. O'Reilly Media Publication, 6th Edition, 2021 (E-Book)
RB-4	Accomazzo Anthony, Murray Nathaniel, Lerner Ari, “ Fullstack React: The Complete Guide to ReactJS and Friends”, Fullstack.io, 2017.

CO-PO MAPPING:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	1	1	1	3	1	3	2	1
CO2	3	2	2	1	3	-	2	2	2

C03	3	3	3	2	1	-	1	2	-
C04	3	3	3	3	2	-	2	2	-
C05	3	3	3	3	1	1	1	2	-